

Wages vs Amenities

Tate Mason

Tate.Mason@uga.edu

John Munro Godfrey Sr. Department of Economics
The University of Georgia

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Outline

- 1 Introduction
- 2 Data
- 3 Empirical Facts
- 4 Model
- 5 Next Steps

Motivation

- Typically, location choice models focus on wages.
- Do other factors play a role in location decisions later in life?
- **Main Question:** How does motivation for moving change with age?

Data Source

- IPUMS USA data from 1990 - 2019
- Decennial censuses and ACS data
- Panel structure by linking individuals across years

Sample Selection

- Individuals aged 40-85 at time of first observation
- Moved at least once during observation period
- Exclude group quarters, military, and missing data
- Final sample size: 1.25 million households

Empirical Motivation

- Younger movers and older movers exhibit different mobility patterns
- High income individuals move more with higher frequency
- Those without children move more often
- Probability of moving peaks at the age around retirement, and then declines

Summary Statistics

Table: Summary Statistics: Movers vs. Non-Movers

	Movers	Non-Movers
N (households)	185,813	940,861
% own home	42.4%	37.6%
% married	51.2%	38.4%
Avg. household income (\$)	54,230.81	42,473.42
Avg. rent (renters / non-missing, \$)	595.87	574.68
Avg. home value (owners / non-missing, \$)	290,351	270,827

Graphical Evidence

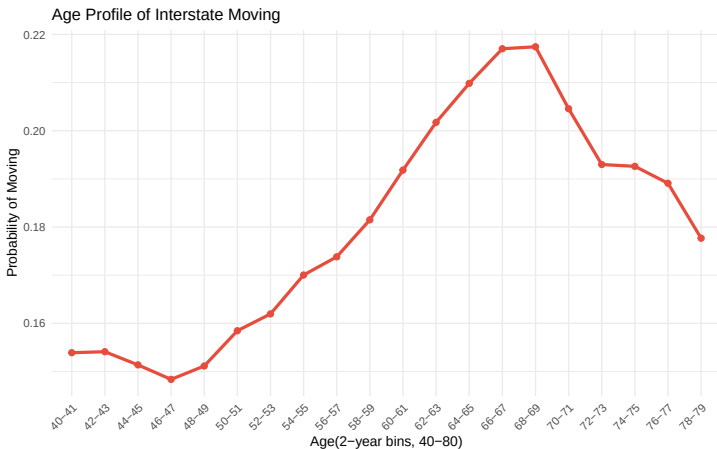


Figure: Age Profile of Moving Frequency

Graphical Evidence

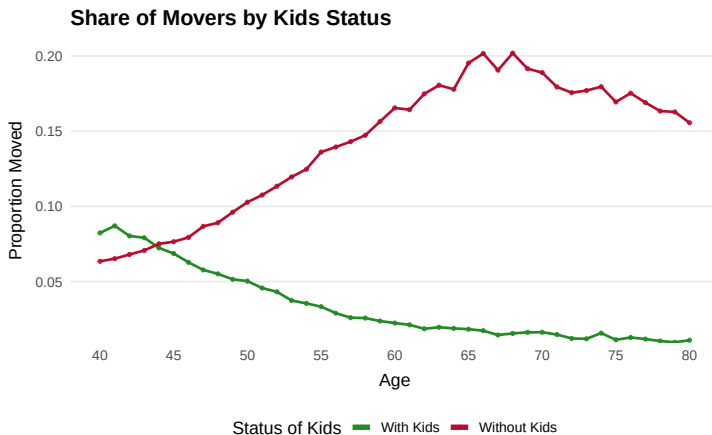


Figure: Children vs No Children Moving Frequency

Graphical Evidence

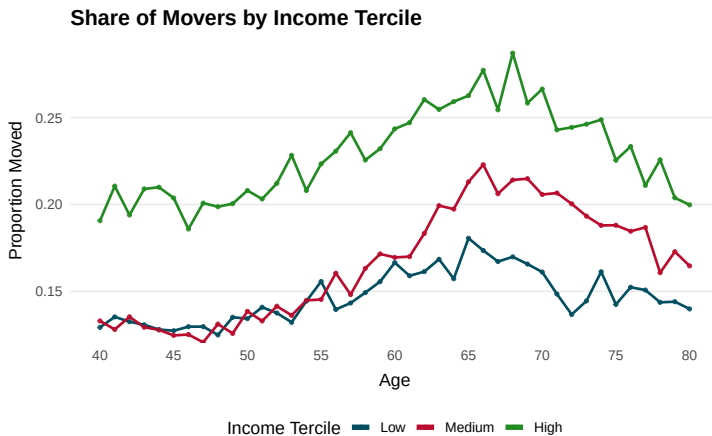


Figure: Income vs Moving Frequency

Graphical Evidence

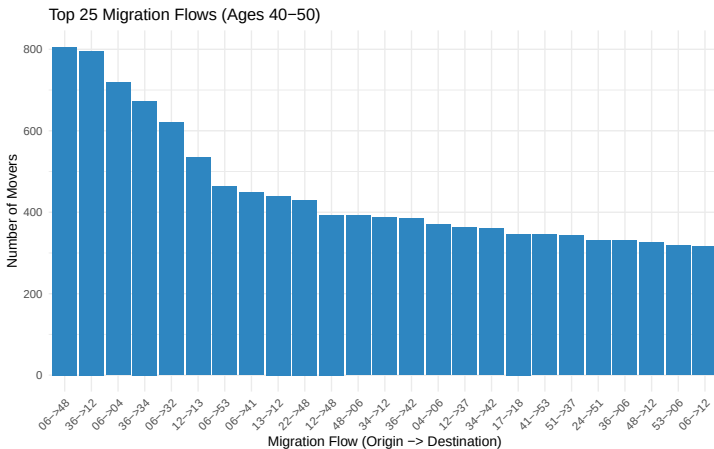


Figure: Migration Flows (Age 40-50)

Graphical Evidence

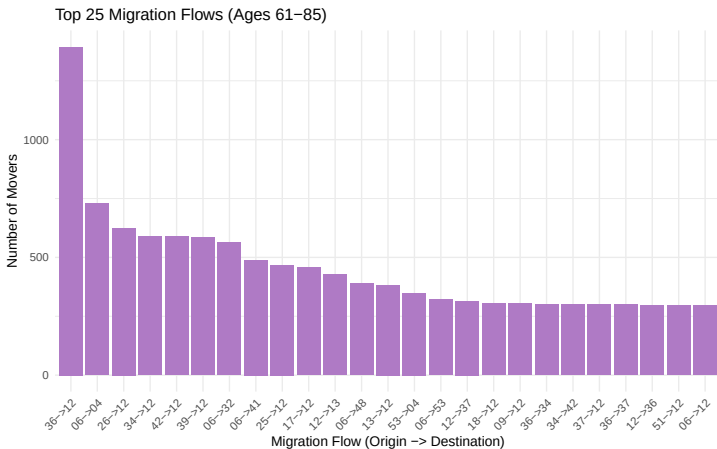


Figure: Migration Flows (Age 61-85)

Model Overview

- Two types: Workers, Retirees
- Workers supply labor inelastically, earn wages based on location, retire with probability $1 - x$ each period
- Retirees receive pension income, base moving decisions on amenities, die with probability $1 - y$ each period, replaced by new workers
- Both types maximize lifetime utility over consumption and amenities

Utility Function

- Workers: $U^w(c, m(z); t) = \frac{-(c-\bar{c})^2}{2} - \gamma \cdot \{z' \neq z\}$
- Retirees: $U^r(c, m(z); t) = \frac{-(c-\bar{c})^2}{2} - \gamma \cdot \{z' \neq z\}$
- c : consumption, $m(z)$: amenities in location z
- $h(z)$: housing quality for workers, $hc(z)$: health care quality for retirees
- γ : moving cost parameter

Budget Constraints

- Workers: $c_t = w(z_t) - \tau(z_t) - \mathbb{E}[c(h(z))|\delta]$
- δ : Prior assumptions about housing costs in location z'
- $w(z_t)$: wage in location z_t , $\tau(z_t)$: taxes, $c(h(z))$: housing cost
- Retirees: $c_t + s_{t+1} = (1 + r)s_t + p - \mathbb{E}[c(h(z))|\delta]$
- p : pension income

Decision Problem: Workers

$$V^w(\delta, z_t, t) = \max\{V_w^M(\delta, z_t, t), V_w^S(\delta, z_t, t)\} \quad (1)$$

- V_w^M : value of moving, V_w^S : value of staying

$$V_w^S(\delta, z_t, t) = \max_{c_t} \{U(c_t) + x\beta V^w(\delta, z_t, t+1) + (1-x)r(\delta, z_t, t+1)\} \quad (2)$$

$$V_w^M(s_t, z_t; t) = \max_{z', c_t} \{U(c_t, m(z')) + x\beta \mathbb{E}[V^w(\delta', z'; t+1)|\delta] + (1-x)\beta \mathbb{E}[V^r(\delta', z'; t+1)|\delta]\} \quad (3)$$

Decision Problem, Retirees

$$V^r(\delta, z_t; t) = \max\{V_r^M(\delta, z_t; t), V_r^S(z_t; t)\} \quad (4)$$

- V_r^M : value of moving, V_r^S : value of staying

$$V_r^S(z_t; t) = \max_{c_t} \{U(c_t, m(z_t)) + y\beta V^r(\delta, z_t; t + 1)\} \quad (5)$$

$$V_r^M(\delta, z_t; t) = \max_{z', c_t} \{U(c_t, m(z')) + y\beta V^r(\delta', z'; t + 1)\} \quad (6)$$

Next Steps

- Find a suitable measure of quality of life for locations
 - Health care quality
 - Climate
 - Cost of Living
- Calibrate model parameters using observed mobility patterns

Thank You

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Questions?